

HLD101 · SST-2028 · PROJECT REPORT

Search Typeahead System

A low-latency search-suggestion service: prefix typeahead, a distributed cache routed by consistent hashing, batched aggregated writes, and recency-aware trending.

Python (standard library only)

SQLite

Vanilla JS

Trie index

Consistent hashing

Batch writes

Repository	<code>github.com/rkawasthiii/HLD-college-project-</code>
Commit for grading	full 40-char hash submitted on the form
Run command	<code>python scripts/generate_dataset.py && python -m backend.server</code>
Dataset size	120,000 queries (Zipf-distributed; $\geq 100k$ required)

1. Architecture

The system is a single Python process that serves both the JSON API and the UI. Reads are served **cache-first** from an in-memory Trie; writes are **buffered and flushed in aggregate** to SQLite. Every concept is implemented from scratch (no frameworks) so each design decision can be explained directly.

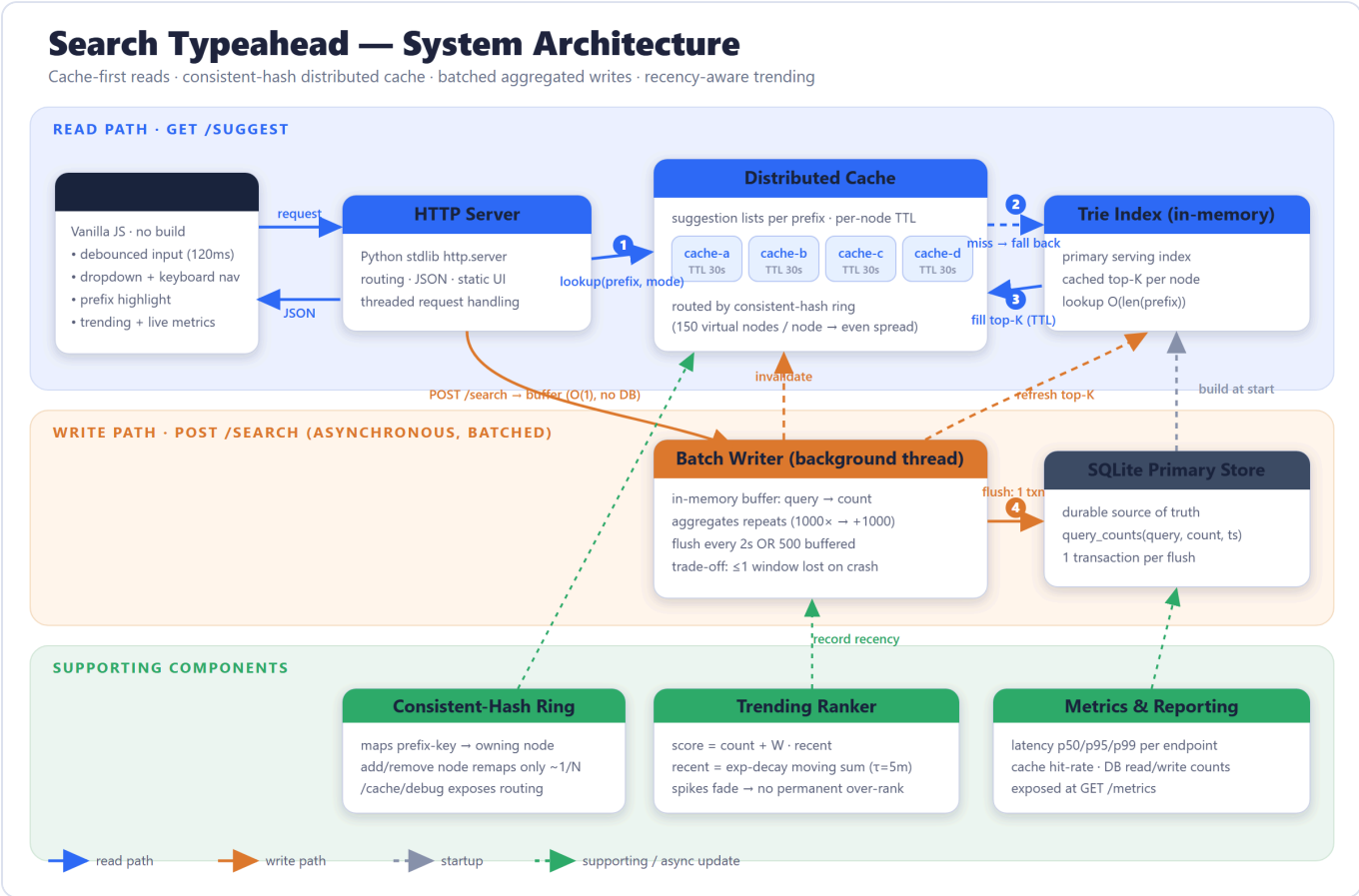


Figure 1 — Read path (blue), write path (orange), and supporting components. Numbered steps ①–④ trace a suggestion lookup and a search submission.

Components

Component	File	Responsibility
HTTP Server	backend/server.py	Routing, JSON IO, static hosting; threaded.
Trie Index	backend/trie.py	In-memory serving index; cached top-K per node.
Distributed Cache	backend/cache.py	4 logical nodes, each a TTL dict; hit/miss counters.
Consistent-Hash Ring	backend/consistent_hash.py	Maps a prefix key to its owning cache node.
Batch Writer	backend/batch_writer.py	Buffers submissions; flushes aggregated writes.
Primary Store	backend/store.py	Durable query_counts table in SQLite.
Trending Ranker	backend/trending.py	Time-decayed recency score for ranking.
Metrics	backend/metrics.py	Latency percentiles, hit-rate, DB op counters.

Read path — GET /suggest?q=<prefix>

The prefix and mode form a cache key "{mode}:{prefix}". The consistent-hash ring selects the owning cache node ①. On a **hit** the cached suggestion list is returned immediately. On a **miss** ② the system falls back to the Trie (the in-memory primary index), which returns the cached top-K for that prefix in $O(\text{len}(\text{prefix}))$; the result fills the cache with a TTL ③ and is returned.

Write path — POST /search

A submission is **not** written to the database inline. It adds +1 to an in-memory buffer and returns {"message": "Searched"} in $O(1)$. A background thread flushes the buffer ④ every 2 seconds or once 500 distinct queries accumulate: it aggregates repeats, writes them in a single SQLite transaction, refreshes the Trie's top-K, feeds the recency tracker, and invalidates the affected cached prefixes.

2. Dataset — source & loading

Format & size

The store ingests a plain-text file, one record per line, as `query<TAB>count` (commas are also accepted). The project ships a generator that produces **120,000** distinct queries (the assignment requires $\geq 100,000$).

```
iphone          1000001
samsung         500000
iphone charger  18184
java tutorial    2078
best laptop review 64
```

Why a synthetic dataset

Counts follow a **Zipf-like (power-law)** distribution — a few head queries are extremely popular and a long tail is rare — which is the real shape of search traffic and the shape that makes typeahead and caching meaningful. Generating it guarantees the project runs out-of-the-box at the required size with no download or licensing friction. Queries are composed from realistic brands, products, tech topics, adjectives, and intent tails (e.g. *"iphone charger"*, *"best laptop review"*, *"java interview questions"*), with short clean queries ranked highest so the typeahead head looks natural.

Loading instructions

```
# 1. generate the dataset (writes data/queries.txt)
python scripts/generate_dataset.py          # default 120,000
python scripts/generate_dataset.py 250000  # optional: larger

# 2. start the server — first run ingests the file into SQLite, then builds the Trie
python -m backend.server
```

Ingestion is idempotent and runs only when the database is empty. On startup the log reports rows ingested and Trie build time.

Swapping in a real open-source dataset

Produce the same `query<TAB>count` format, save it as `data/queries.txt`, delete `data/typeahead.db*`, and restart. Suitable sources: **Wikipedia page titles** (use pageviews as the count), the **AOL query log** (aggregate identical queries), or any **e-commerce product catalogue** (use views/sales). If a source has no counts, derive them by counting duplicate occurrences.

3. API documentation

Base URL `http://127.0.0.1:8000`. All responses are JSON. A machine-readable summary is served at `GET /api`.

GET `/suggest?q=<prefix>&mode=count|trending&limit=10`

Up to 10 prefix-matching suggestions. `mode=count` (default) ranks by total search count; `mode=trending` ranks by the recency-aware score.

```
// GET /suggest?q=iphone
{ "prefix": "iphone", "mode": "count", "source": "cache",
  "suggestions": [ { "query": "iphone", "count": 1000001 },
                   { "query": "iphone charger", "count": 18184 } ] }
```

POST `/search`

Body `{ "query": "... " }`. Records the query via the batch buffer and returns a dummy response. Does not touch the database inline.

```
// POST /search { "query": "iphone 15 pro" }
{ "message": "Searched", "query": "iphone 15 pro", "recorded": true }
```

GET `/trending?limit=10`

Top queries overall by the recency-aware trending score, each with its real count, decayed recent activity, and combined score.

GET `/cache/debug?prefix=<p>&mode=count`

Explains routing: which cache node owns the prefix (consistent hashing), HIT/MISS state, the key & ring-point hashes, and a sample of how keys distribute across nodes.

```
// GET /cache/debug?prefix=iphone
{ "prefix": "iphone", "owner_node": "cache-a", "state": "HIT",
  "sample_distribution": { "cache-a": 6, "cache-b": 2, "cache-c": 3, "cache-d": 4 },
  "ring_nodes": [ "cache-a", "cache-b", "cache-c", "cache-d" ], "vnodes_per_node": 150 }
```

GET `/metrics`

Latency p50/p95/p99 per endpoint, cache hit-rate (overall and per node), batch write-reduction, DB op counts, and Trie size. `GET /healthz` is a liveness probe.

Method & path	Purpose
GET /suggest	Prefix suggestions (count or trending).
POST /search	Record a search (batched); returns "Searched".
GET /trending	Recency-aware trending list.
GET /cache/debug	Consistent-hash routing + HIT/MISS.

GET /metrics	Latency, hit-rate, write-reduction.
GET /healthz	Liveness.

4. Design choices & trade-offs

Trie with cached top-K (vs. scanning a sorted list)

Each Trie node caches the best K completions of its prefix, so a lookup is $O(\text{len}(\text{prefix}))$ with no subtree scan. The top-K is maintained incrementally and is **exact** because search counts only ever increase: a query can only *enter* a node's top-K by growing, which the update path always catches. A plain sorted-array scan would re-sort thousands of completions for short, hot prefixes on every miss.

Consistent hashing with virtual nodes

The cache is sharded across 4 logical nodes. Each is placed at 150 points on a hash ring, which (a) spreads keys evenly — measured $\approx 25\%$ per node — and (b) means adding/removing a node remaps only $\sim 1/N$ of keys instead of the whole keyspace. MD5 is used purely as a fast, well-distributed hash, not for security.

Trade-off: more virtual nodes give a smoother distribution but a larger ring and slightly slower lookups. 150 is a good balance for 4 nodes.

Cache TTL + active invalidation

Cached suggestion lists expire after a 30-second TTL as a safety net, and a batch flush also explicitly deletes the prefixes whose counts just changed. This keeps suggestions fresh without waiting for the TTL, while the TTL bounds staleness for anything not actively invalidated.

Batch writes (the core write-reduction mechanism)

POST /search is $O(1)$ and never touches SQLite; it only increments an in-memory buffer. A background thread flushes every 2 s or every 500 buffered queries, aggregating repeats so 1000 searches of "iphone" become a single +1000 upsert in one transaction. This collapses thousands of writes into a handful.

Failure trade-off: submissions buffered but not yet flushed are lost if the process crashes. The exposure is **bounded** by the flush interval/size ($\leq$ one window). We accept this small, bounded durability gap in exchange for a large reduction in write amplification. A write-ahead log on the buffer would close the gap at the cost of added complexity.

Recency-aware trending

The same /suggest API serves both rankings. **Basic** (mode=count) ranks by total count. **Enhanced** (mode=trending) uses $\text{score} = \text{count} + W \cdot \text{recent}$, where recent is an exponentially-decaying moving sum updated on each flush:

```
recent ← recent · exp(-Δt / τ) + delta    # τ = 5 min
score  ← count + W · recent               # W = 20,000
```

Why it can't over-rank forever: because the recency term decays exponentially, a query that is hot for a short window loses its boost within a few τ and falls back to its base count ranking. This directly satisfies the requirement to avoid permanently over-ranking a briefly popular query. The hot-set is pruned once a query's decayed value approaches zero, keeping the structure small.

Why standard-library Python + SQLite

No `pip install` means the project runs anywhere with just Python, which satisfies "easy to run locally" and keeps every mechanism visible and explainable. SQLite is a real, file-based database with transactional writes — ideal for demonstrating batched DB writes without running a separate server.

5. Performance report

Measured on Windows 11, Python 3.14, 120,000-query dataset, via `scripts/benchmark.py` (drives the running server over HTTP). All numbers are reproducible by running that script.

0.08 ms

SUGGEST P95 (SERVER)

~92%

CACHE HIT RATE

666×

WRITE REDUCTION

120k

INDEXED QUERIES

Suggestion latency

Measurement	p50	p95	Notes
Server-side /suggest	~0.05 ms	0.08 ms	internal timing (cache + Trie)
End-to-end (HTTP, localhost)	~1.9 ms	~2.3 ms	includes loopback + JSON overhead
Cold (forced cache miss)	~1.9 ms	~2.1 ms	Trie lookup still O(len(prefix))

The Trie keeps even cold lookups fast, and the distributed cache absorbs repeated hot prefixes — measured hit rate $\approx 92\%$ under the benchmark's repeated-prefix workload.

Batch-write reduction

Metric	Value
Search submissions in test burst	2,000
SQLite write transactions used	3
Reduction for the burst	~666× fewer writes
Steady-state overall reduction (submissions ÷ txns)	~125×–195×

Because every submission is buffered and only flushed in aggregate, write amplification drops by two to three orders of magnitude versus one-write-per-search.

Consistent-hash distribution

A sample of prefix keys routed across the 4 cache nodes: `cache-a:6`, `cache-b:2`, `cache-c:3`, `cache-d:4` — close to even given the small sample, and exposed live at `/cache/debug`. Unit tests assert each node receives 15–35% of 8,000 keys and that removing a node remaps <45% of keys.

Recency-aware trending — before/after

For prefix `"co"`, the tail query `"consistent hashing"` has a base count of ~650. After 1,500 recent searches its trending score rose to **~29,660,989** (count + `W-recent`), dominating the ranking; with no further activity the boost decays back toward the base count over a few τ . This demonstrates both the lift from recent activity and the anti-over-ranking decay.

Reproduce

```
# terminal 1
python -m backend.server
# terminal 2
python scripts/benchmark.py      # prints latency, hit-rate, write reduction, trending
python -m unittest discover -s tests -v  # 8 component tests
```